Chapter 4. Systematic Sampling

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1 Introduction

To select n names from a long list, a simple way is to pick names at equal intervals, e.g. every tenth name. If the starting point is random, the result is a systematic sample.

Definition 4.1

A **1-in-k systematic sample with a random start** is obtained by randomly selecting one element among the first k elements of the frame and every kth element thereafter. Here n denotes the sample size and k the number of possible systematic samples.

Advantages over SRS:

- the sample is easy to draw and the field work is less prone to mistakes;
- it can provide more information per unit cost, because the sample is spread evenly over the frame.

2 Selection Methods

2.1 Method A: random start among the first k

Example 4.2: $N = 12$, $k = 3$

Pick a random unit among y_1, y_2, y_3 and then every third unit. If y_2 is picked, the sample is y_2, y_5, y_8, y_{11}. The population is cut into $n = N/k = 4$ rows and one unit is taken from each row; the $k = 3$ possible samples are the columns

#1	#2	#3
y_1	y_2	y_3
y_4	y_5	y_6
y_7	y_8	y_9
y_{10}	y_{11}	y_{12}

In general the sample is $j,\ j+k,\ j+2k,\ \ldots,\ j+(n-1)k$ with j uniform on $\{1,\ldots,k\}$, and each systematic sample has probability $1/k$.

If $N \neq nk$, the samples have unequal sizes. For $N = 12$ and $k = 5$ we have $2 < N/k = 2.4 < 3$: the five samples are (y_1, y_6, y_{11}), (y_2, y_7, y_{12}), (y_3, y_8), (y_4, y_9), (y_5, y_{10}), of size 2 or 3, yet each is chosen with probability $1/5$.

2.2 Method B: random start among all N

Select one unit, say the jth, at random from the whole population, and divide j by k: if the remainder is r, the start is y_r (with $r = 0$ meaning y_k). Then take every kth unit.

Example 4.3: $N = 11$, $k = 3$

If $j = 8$, then $8 = 2 \cdot 3 + 2$, so $r = 2$, the start is y_2 and the sample is (y_2, y_5, y_8, y_{11}). The possible samples are

$$\#1 = (y_1, y_4, y_7, y_{10}), \quad \#2 = (y_2, y_5, y_8, y_{11}), \quad \#3 = (y_3, y_6, y_9).$$

Sample #2 is obtained whenever $j \in \{2, 5, 8, 11\}$, so its probability is $4/11$; similarly #1 has probability $4/11$ and #3 has $3/11$.

Remark 4.4: A versus B

Both methods produce the same collection of systematic samples, but the selection probabilities differ: $1/k$ under Method A and n_i/N (proportional to the sample size) under Method B. Method B requires N to be known. When $N = nk$ the two coincide.

3 Estimation of the Population Mean

Let y_{ij} be the jth element of the ith systematic sample ($i = 1, \ldots, k$, $j = 1, \ldots, n_i$), $\bar{y}_i = \frac{1}{n_i}\sum_j y_{ij}$, and $\bar{y}_{\text{sy}}$ the mean of the sample actually selected.

3.1 Unbiasedness

Case 1 (Method A, $N = nk$).

$$\mathrm{E}(\bar{y}_{\mathrm{sy}}) = \frac{1}{k}(\bar{y}_1 + \cdots + \bar{y}_k) = \frac{1}{k}\frac{1}{n}(y_1 + \cdots + y_N) = \mu,$$

so $\bar{y}_{\mathrm{sy}}$ is unbiased.

Example 4.5

Population $6, 3, 4, 9, 2, 5, 1, 7, 8$ with $k = 3$: the samples are $(6, 9, 1)$, $(3, 2, 7)$, $(4, 5, 8)$ with means $16/3$, $12/3$, $17/3$, whose average is $5 = \mu$.

Case 2 (Method A, $N \neq nk$). For $y_1, \ldots, y_7$ and $k = 3$,

$$\mathrm{E}(\bar{y}_{\mathrm{sy}}) = \frac{1}{3}\left[\frac{y_1 + y_4 + y_7}{3} + \frac{y_2 + y_5}{2} + \frac{y_3 + y_6}{2}\right] \neq \frac{y_1 + \cdots + y_7}{7} = \mu$$

in general: $\bar{y}_{\mathrm{sy}}$ is *biased*.

Case 3 (Method B, $N \neq nk$). With probabilities $3/7, 2/7, 2/7$,

$$\mathrm{E}(\bar{y}_{\mathrm{sy}}) = \frac{3}{7}\cdot\frac{y_1 + y_4 + y_7}{3} + \frac{2}{7}\cdot\frac{y_2 + y_5}{2} + \frac{2}{7}\cdot\frac{y_3 + y_6}{2} = \frac{y_1 + \cdots + y_7}{7} = \mu.$$

In general $\mathrm{E}(\bar{y}_{\mathrm{sy}}) = \sum_i \frac{n_i}{N}\bar{y}_i = \frac{1}{N}\sum_i \sum_j y_{ij} = \mu$: Method B is always unbiased.

3.2 Variance of $\bar{y}_{\mathrm{sy}}$ ($N = nk$)

By definition,

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{k}\sum_{i=1}^{k}(\bar{y}_i - \mu)^2,$$

where $S^2 = \frac{1}{N-1}\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \mu)^2$.

Theorem 4.6: Variance of the systematic mean

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{N-1}{N}S^2 - \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_i)^2.$$

Proof. Decompose the total sum of squares:

$$\sum_i\sum_j (y_{ij} - \mu)^2 = \sum_i\sum_j (y_{ij} - \bar{y}_i)^2 + n\sum_i(\bar{y}_i - \mu)^2 + 2\sum_i(\bar{y}_i - \mu)\sum_j(y_{ij} - \bar{y}_i),$$

and the cross term vanishes because $\sum_j(y_{ij} - \bar{y}_i) = 0$. The left side is $(N-1)S^2$ and $n\sum_i(\bar{y}_i - \mu)^2 = nk\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = N\mathrm{Var}(\bar{y}_{\mathrm{sy}})$. Divide by N. ∎

The first term is the variance of the population as a whole; the second is the pooled variation *within* the k systematic samples. The more heterogeneous the units within a systematic sample, the smaller $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$ becomes.

Remark 4.7: Periodicity

For the population 1 2 3 4 5 | 1 2 3 4 5 | 1 2 3 4 5 with $k = 5$, every systematic sample is constant, e.g. $(2, 2, 2)$. There is no within-sample variation, so $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$ is as large as it can be.

3.3 Intraclass correlation

Homogeneity within a systematic sample is measured by the **intraclass correlation coefficient**

$$\rho = \frac{\mathrm{E}[(y_{ij} - \mu)(y_{ij'} - \mu)]}{\mathrm{E}[(y_{ij} - \mu)^2]}, \qquad j \neq j',$$

where the expectations are averages over the $kn(n-1)/2$ pairs in the same sample and over the N units, respectively.

Theorem 4.8

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{S^2}{n} \cdot \frac{N-1}{N}[1 + (n-1)\rho], \qquad \rho = \frac{2}{(n-1)(N-1)S^2} \sum_{i=1}^{k} \sum_{j<j'} (y_{ij} - \mu)(y_{ij'} - \mu).$$

Derivation

$$\begin{aligned}\mathrm{Var}(\bar{y}_{\mathrm{sy}}) &= \frac{1}{k} \sum_i \Big(\frac{1}{n} \sum_j y_{ij} - \mu \Big)^2 = \frac{1}{n^2 k} \sum_i \Big(\sum_j (y_{ij} - \mu) \Big)^2 \\ &= \frac{1}{n^2 k} \Big[\sum_i \sum_j (y_{ij} - \mu)^2 + 2 \sum_i \sum_{j<j'} (y_{ij} - \mu)(y_{ij'} - \mu) \Big].\end{aligned}$$

The denominator of ρ is $\frac{1}{N} \sum_i \sum_j (y_{ij} - \mu)^2 = \frac{N-1}{N} S^2$, and the numerator is $\frac{2}{kn(n-1)} \sum_i \sum_{j<j'} (\cdot)(\cdot)$, so

$$2 \sum_i \sum_{j<j'} (y_{ij} - \mu)(y_{ij'} - \mu) = kn(n-1) \frac{N-1}{N} S^2 \rho = (n-1)(N-1) S^2 \rho,$$

which is the stated formula for ρ. Substituting, with $n^2 k = nN$,

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{nN}[(N-1)S^2 + (n-1)(N-1)S^2\rho] = \frac{S^2}{n} \frac{N-1}{N}[1 + (n-1)\rho].$$

Compare with SRS, $\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{S^2}{n}(1 - \frac{n}{N})$:

- ρ large and positive (homogeneous systematic samples) $\Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}})$ large;
- ρ small positive or negative (heterogeneous samples) $\Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}})$ small;
- $\rho \approx 0 \Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}}) \approx \mathrm{Var}(\bar{y}_{\mathrm{ran}})$. Indeed $\rho = -\frac{1}{N-1}$ reproduces the SRS variance exactly.

3.4 ANOVA form

Arrange the population as an $n \times k$ table whose columns are the systematic samples, with column means $\bar{y}_1, \ldots, \bar{y}_k$ and grand mean μ. Let

$$SS_T = \sum_i \sum_j (y_{ij} - \mu)^2, \quad SS_W = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2, \quad SS_B = n \sum_i (\bar{y}_i - \mu)^2,$$

so $SS_T = SS_W + SS_B$ and $S^2 = SS_T/(N-1)$. Then

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{k}\sum_i (\bar{y}_i - \mu)^2 = \frac{SS_B}{N},$$

and equating with the ρ form, $\frac{SS_T}{nN}[1+(n-1)\rho] = \frac{SS_B}{N}$, gives

$$1+(n-1)\rho = \frac{n\,SS_B}{SS_T} \quad\Longrightarrow\quad \rho = 1 - \frac{n}{n-1}\frac{SS_W}{SS_T} \approx 1 - \frac{SS_W}{SS_T} = \frac{SS_B}{SS_T}.$$

A large between-sample sum of squares means a large ρ and a large variance.

Example 4.9: Two orderings of $1,\ldots,9$

Here $\mu = 5$, $S^2 = 60/8 = 7.5$, $n = k = 3$, and the SRS variance is $\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{7.5}{3}\cdot\frac{6}{9} = 1.667$.

- Order 1 2 3 | 4 5 6 | 7 8 9: the samples are $(1,4,7),(2,5,8),(3,6,9)$ with means $4,5,6$. Then $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{3}(1+0+1) = \frac{2}{3}$ and $\rho = -21/60$. Heterogeneous samples: systematic beats SRS.
- Order 1 4 7 | 2 5 8 | 3 6 9: the samples are $(1,2,3),(4,5,6),(7,8,9)$ with means $2,5,8$. Then $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{3}(9+0+9) = 6$ and $\rho = 51/60$. Homogeneous samples: systematic is much worse than SRS.

Check: $\frac{7.5}{3}\cdot\frac{8}{9}(1+2\rho) = \frac{20}{9}(1+2\rho)$ equals $\frac{2}{3}$ for $\rho = -0.35$ and 6 for $\rho = 0.85$.

3.5 Types of population

We want systematic samples with heterogeneous units (small ρ). Which populations produce them?

1. **Random order.** E.g. estimating the average weight of $N = 3000$ SNU freshmen from an alphabetical list. The units in a systematic sample are effectively in random order, $\rho \approx 0$, and $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) \approx \mathrm{Var}(\bar{y}_{\mathrm{ran}})$: the systematic sample may be treated as an SRS.
2. **Ordered population.** E.g. farms ordered by area when estimating corn yield. A systematic sample covers the whole range, is heterogeneous, and $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) < \mathrm{Var}(\bar{y}_{\mathrm{ran}})$. It avoids samples with too many large or too many small farms, so it tends to be more representative than an SRS.
3. **Periodic variation.** E.g. supermarket sales, high on Friday–Saturday and low on Monday–Tuesday. If k matches the period, the units are homogeneous, ρ is large, and $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) > \mathrm{Var}(\bar{y}_{\mathrm{ran}})$.

3.6 Estimating the variance

A single systematic sample contains no information about between-sample variation, so $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$ cannot be estimated unbiasedly from it. Assuming a randomly ordered population ($\rho \approx 0$), we use the SRS formula:

$$\bar{y}_{\mathrm{sy}} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \widehat{\mathrm{Var}}(\bar{y}_{\mathrm{sy}}) = \frac{s^2}{n}\cdot\frac{N-n}{N}.$$

This is conservative for an ordered population and anti-conservative under periodicity.

4 Population Total and Proportion

Assuming a randomly ordered population:

$$\hat{\tau} = N\bar{y}_{\text{sy}}, \qquad \widehat{\text{Var}}(\hat{\tau}) = N^2\widehat{\text{Var}}(\bar{y}_{\text{sy}}) = N^2\frac{s^2}{n}\frac{N-n}{N};$$

$$\hat{p}_{\text{sy}} = \bar{y}_{\text{sy}} = \frac{1}{n}\sum y_i \ (y_i \in \{0,1\}), \qquad \widehat{\text{Var}}(\hat{p}_{\text{sy}}) = \frac{\hat{p}_{\text{sy}}\hat{q}_{\text{sy}}}{n-1}\cdot\frac{N-n}{N}.$$

5 Selecting the Sample Size

Under the same assumption the SRS formulas apply: for μ with bound B,

$$n = \frac{N\sigma^2}{(N-1)D+\sigma^2} = \frac{NS^2}{ND+S^2}, \qquad D = \frac{B^2}{4},$$

and for p, $n = \dfrac{Npq}{(N-1)D+pq}$ with $D = B^2/4$.

6 Selected exercises

Motivation. The first item combines systematic selection with stratification; the second quantifies exactly how much a systematic sample gains in an ordered population, the case described qualitatively above; the third is a computation for a total under the random-order assumption.

Exercise 4.1 (Systematic sampling within strata)

A population of N elements is divided into L strata of sizes $N_\ell = n_\ell k$, $\ell = 1,\dots,L$, with the same k in every stratum. In each stratum a 1-in-k systematic sample with a random start is drawn (Method A), independently across strata. Let $\bar{y}_\ell$ be the mean of the sample from stratum ℓ and $\hat{\mu} = \sum_\ell w_\ell \bar{y}_\ell$, $w_\ell = N_\ell/N$. Find $\text{E}(\hat{\mu})$ and $\text{Var}(\hat{\mu})$.

Solution. Write $y_{\ell ij}$ for the jth element of the ith possible systematic sample in stratum ℓ, $\bar{y}_{\ell i} = \frac{1}{n_\ell}\sum_j y_{\ell ij}$, and μ_ℓ, S_ℓ^2, ρ_ℓ for the stratum mean, variance and intraclass correlation.

Step 1. (Unbiasedness.) In stratum ℓ the sample i is chosen with probability $1/k$, so by Case 1 of the unbiasedness subsection, $\text{E}(\bar{y}_\ell) = \frac{1}{k}\sum_i \bar{y}_{\ell i} = \frac{1}{N_\ell}\sum_i\sum_j y_{\ell ij} = \mu_\ell$. Hence $\text{E}(\hat{\mu}) = \sum_\ell w_\ell\mu_\ell = \frac{1}{N}\sum_\ell \tau_\ell = \mu$.

Step 2. (Variance.) The random starts are independent across strata, so the covariances vanish:

$$\text{Var}(\hat{\mu}) = \sum_{\ell=1}^{L} w_\ell^2 \text{Var}(\bar{y}_\ell), \qquad \text{Var}(\bar{y}_\ell) = \frac{1}{k}\sum_{i=1}^{k}(\bar{y}_{\ell i}-\mu_\ell)^2.$$

Step 3. (Intraclass form.) Applying the intraclass-correlation theorem within stratum ℓ,

$$\text{Var}(\hat{\mu}) = \sum_{\ell=1}^{L} w_\ell^2 \frac{S_\ell^2}{n_\ell}\cdot\frac{N_\ell-1}{N_\ell}\big[1+(n_\ell-1)\rho_\ell\big].$$

Step 4. (Interpretation.) Since $n_\ell = N_\ell/k$, the design is automatically proportionally allocated with common sampling fraction $1/k$. If every stratum is in random order ($\rho_\ell \approx -1/(N_\ell-1)$), the bracket times $\frac{N_\ell-1}{N_\ell}$ is $1-\frac{1}{k}$ and $\text{Var}(\hat{\mu})$ reduces to that of proportional stratified sampling. A

single systematic sample from each stratum still gives no unbiased variance estimator, so in practice $\mathrm{Var}(\bar{y}_\ell)$ is estimated by the SRS formula, stratum by stratum. ■

Exercise 4.2 (Population with a linear trend)

Let $y_t = a + bt$, $t = 1, \dots, N$, with $b \neq 0$ and $N = nk$, $n \geq 2$. Show that

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{b^2(k^2-1)}{12}, \qquad \mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{b^2(k-1)(N+1)}{12}, \qquad \frac{\mathrm{Var}(\bar{y}_{\mathrm{sy}})}{\mathrm{Var}(\bar{y}_{\mathrm{ran}})} = \frac{k+1}{N+1} < 1.$$

Solution. [Supplement]

Step 1. (A discrete uniform variance.) If I is uniform on $\{1, \dots, k\}$, then $\mathrm{E}I = \frac{k+1}{2}$, $\mathrm{E}I^2 = \frac{(k+1)(2k+1)}{6}$ and

$$\mathrm{Var}I = \frac{(k+1)(2k+1)}{6} - \frac{(k+1)^2}{4} = \frac{(k+1)\big[2(2k+1) - 3(k+1)\big]}{12} = \frac{k^2-1}{12}.$$

Step 2. (Systematic mean.) Sample i consists of $t = i, i+k, \dots, i+(n-1)k$, so $\bar{y}_i = a + b\big(i + \frac{(n-1)k}{2}\big)$. The start I is uniform on $\{1, \dots, k\}$, so $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = b^2 \mathrm{Var}I = \frac{b^2(k^2-1)}{12}$.

Step 3. (SRS.) Step 1 with k replaced by N gives $\sum_{t=1}^{N} \big(t - \frac{N+1}{2}\big)^2 = N\frac{N^2-1}{12}$, so $S^2 = \frac{b^2}{N-1} \cdot \frac{N(N^2-1)}{12} = \frac{b^2 N(N+1)}{12}$. With $1 - f = 1 - \frac{1}{k}$ and $N = nk$,

$$\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{k-1}{k} \cdot \frac{1}{n} \cdot \frac{b^2 N(N+1)}{12} = \frac{b^2(k-1)(N+1)}{12}.$$

Step 4. (Ratio.) Dividing, $\frac{(k^2-1)}{(k-1)(N+1)} = \frac{k+1}{N+1}$, which is below 1 because $k < N$ when $n \geq 2$. For $N = 1000$ and $k = 10$ the systematic sample has about 1% of the SRS variance: it cannot pick only small or only large units, which is the "ordered population" case of the text made exact. ■

Exercise 4.3 (Families who rent)

A 1-in-50 systematic sample is drawn from the tax records of a city with $N = 15{,}200$ households to estimate the total number of families who rent. Of the $n = 304$ sampled families, $\sum y_i = 88$ rent ($y_i = 1$ if the family rents). Estimate the total and place a bound on the error, assuming the records are in random order with respect to renting.

Solution.

Step 1. $\hat{p}_{\mathrm{sy}} = 88/304 = 0.2895$ and $\hat{\tau} = N\hat{p}_{\mathrm{sy}} = 15{,}200 \times 0.2895 = 4400$.

Step 2. Under the random-order assumption the SRS formulas apply:

$$\widehat{\mathrm{Var}}(\hat{p}_{\mathrm{sy}}) = \frac{\hat{p}\hat{q}}{n-1} \cdot \frac{N-n}{N} = \frac{0.2895 \times 0.7105}{303} \cdot \frac{14{,}896}{15{,}200} = 0.000665.$$

Step 3. $\widehat{\mathrm{Var}}(\hat{\tau}) = N^2 \widehat{\mathrm{Var}}(\hat{p}_{\mathrm{sy}}) = 15{,}200^2 \times 0.000665 = 153{,}700$ and $2\sqrt{153{,}700} = 784$. The estimate is 4400 ± 784 renting families. The bound is trustworthy only if the list has no periodicity of period 50 related to renting (e.g. apartment blocks listed in groups of 50). ■